ALTR Update Assessments
Every change since December 2025, with its measured impact

> **Scope.** The main model — and the public-facing `altr-model-refactored` — covers the **power sector only**. Cement, steel, coal mining and oil & gas production are out of scope for now; the harness that supports them stays in the code, but no result in this document includes them. Every figure below is power-sector, 25 AR6 providers, 4,883 companies.
>
> **Data provenance (2026-09-01).** Section 9 was re-measured on fully rebuilt inputs: BigQuery marts of 2026-09-01, ownership resolved via `ownership_type=='direct'` (shares now sum to 100% per asset-year; the previous tier filter selected the operator's parent and dropped 68% of assets), the ×100 capacity-allocation bug fixed, and carbon prices corrected at source (they had been frozen at each series' first year). Sections U1–U4 still carry April-2026 isolation-study numbers — see the note on each.

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1 The model as we left it in December 2025

By December the big architectural rework was done: ALTR valued a firm as the sum of its physical power plants. For every asset, each year, in a baseline and a shock scenario:

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EBITDA = revenue (volume × price) − fuel − fixed O&M − carbon cost
FCFF   = EBITDA − CapEx (replacement + new-build + decommissioning)
NPV    = discounted FCFF + terminal value  →  shock = ΔNPV baseline vs
shock
```

What actually drives results, in order of force: (1) the scenario **production trajectories** — how much each asset runs in baseline vs shock; (2) **electricity prices** from the AR6 scenario; (3) the **carbon tax** on emissions; (4) the **discount rate and terminal value**, which translate the cash-flow path into a valuation — and because the terminal value is ~15–20× final-year cash flow, whatever happens in the last modeled years dominates the NPV.

The December settings that matter for everything below: one flat discount rate (7%) for all assets in both scenarios; 2% perpetual terminal growth for every technology — coal plants included; terminal value always a perpetuity; decommissioning recorded as *positive scrap income*; and electricity prices taken raw from the IAM scenarios.

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2 The inconsistency problem

A transition stress test has an economically expected direction: under a below-2°C shock, fossil assets should lose value (negative ΔNPV) and renewables should hold or gain (positive ΔNPV). The December model routinely violated this:

- **Oil was backwards.** Only 28.8% of oil company-technology cells showed the expected negative shock — 71% of oil assets appeared to *benefit* from climate transition. Gas sat at 65%; only coal behaved (92%).
- **Averaged across all 25 usable providers, carbontech direction was 68.5%** — and on partial-equilibrium providers (COFFEE, GCAM) it was a coin flip (~51–52%).
- **Baselines were broken before any shock.** AR6 electricity prices are LCOE-derived and technology-differentiated, not market-clearing — some technologies were perpetually loss-making in the *baseline*, so a shock against that baseline measured nothing.
- **The valuation couldn't tell brown from green.** Flat discounting and uniform 2% perpetual growth valued a declining coal plant with the same machinery as a growing solar farm — at r=7%, g=2% vs g=0% is a 20.4× vs 14.6× terminal multiple, so the uniform assumption overvalued declining fossil by ~28% at the terminal.
- **Retirement paid.** With decommissioning as scrap income, retiring early in a shock *added* cash — muting exactly the losses the test is supposed to measure.

The updates below are the response. Each one is described with its source and intuition, and — where an isolation run exists — its measured differential impact on direction accuracy.

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3 Why we could not run "all" AR6 scenarios

The AR6 database holds over a thousand scenario submissions, but the model has hard input requirements, and each requirement cuts the usable set:

1. **A matched pair.** Every run needs a baseline-like scenario (current policies / no policy) *and* a target-like scenario (~500 Gt carbon budget or net-zero) *from the same provider* — e.g. `EN_NoPolicy` vs `EN_NPi2020_500`. Pairing is automated (`discover_scenario_pairs.py`) with scored heuristics.
2. **Power-sector variables.** The pair must carry capacity/production trajectories, electricity prices, and ideally carbon prices for the power technologies we hold assets in (wind coverage is tracked explicitly).
3. **Overlapping geographies and enough data.** Baseline and target must share regions (the model filters to common geographies), and providers under ~500 rows per scenario are dropped.

After those filters, **25 providers** survive — that is the "all scenarios" universe, and the April isolation study ran every one of them (125 runs). The **MCPR v2 batch used only 5 of the 25** (COFFEE, GCAM 5.2, IMAGE 3.0, MESSAGEix-GLOBIOM, WITCH) for a different reason: each pipeline run takes 5–8 minutes, and MCPR v2 was in an edit-run-fix loop where a full 25-provider × 3-config sweep (~8 hours) per iteration was not practical. The 5 were chosen to span IAM philosophies (partial-equilibrium, energy-system, hybrid). The full-provider rerun is the queued canonical rebuild (decision OP2).

U0 From one provider to twenty-five — the measurement apparatus

`commit 13707f2 (2025-09)` `commit 49efb56 (2026-04)` `scenario_manifest.json`

The issue. Through most of 2025 the model could run on exactly one scenario provider — WITCH — because three things pinned it there: a single hardcoded scenario pair in the config, one hand-enriched data extract (the raw scenario table lacked the fuel, CapEx, O&M and efficiency columns the cash-flow model needs), and a fatal assertion that required baseline and target scenarios to cover *identical* geographies. Most AR6 providers report baseline and target at different regional breakdowns, so even with data they crashed on arrival. One provider means no robustness evidence: every result was conditional on WITCH's modeling philosophy.

What changed. Three steps. (1) Sept 2025: the geography assertion became an intersection — providers now run on the regions their scenario pairs share, with a warning listing what was dropped. (2) A cost-enriched all-provider AR6 extract with pre-computed viability flags replaced the hand-built WITCH file. (3) April 2026: automated pair discovery (`discover_scenario_pairs.py` scores every scenario name for baseline-likeness, carbon budget, and data sufficiency) plus a batch runner that writes per-provider config overrides and runs Kedro unattended — one command instead of hand-edited YAML per provider.

Differential impact. Coverage, not direction:

	BEFORE	AFTER
Scenario providers runnable	1 (WITCH)	25
Runs per comparison campaign	1, manual	125 (isolation study) / 30 (single-vintage rebuild), unattended

Why it leads this document: every measured impact in U1–U7 exists because of this apparatus — no other update could have been *tested* without it. Honest limit: 25 is not "all of AR6"; it is every provider surviving the input requirements (matched policy pair, power-sector cost variables, overlapping geographies, ≥500 rows).

U1 Technology-differentiated discount spreads (D1)

Vintage: the impact figures in this box are from the April-2026 isolation study, on pre-rebuild data. They have not been re-measured against the 2026-09-01 inputs — that needs an `iso_*` sweep (3 configs × 25 providers, ~4.5h). Direction of effect is expected to hold; magnitudes will shift, since the rebuild lifted the vanilla baseline from 71.9% to 81.5%.

`commit 056d1f6` `config iso_d1`

The issue. The December model discounted every asset at the same flat 7% — a coal plant and a solar farm carried identical risk pricing. Markets demonstrably price carbon exposure into the cost of capital, so a flat rate systematically overvalued brown assets relative to green ones, muting or flipping the very shocks the test is supposed to measure.

What changed. Carbontech assets discounted at +100bps over the 7% base rate, greentech at −50bps, in both baseline and shock. Replaces the flat single rate that priced coal and solar risk identically.

Source & intuition. Bolton & Kacperczyk's carbon premium (high-emission firms face ~1.5–2.5% higher equity returns → ~60–150bps WACC spread); Shell's 2024 annual report discloses 7.5% O&G vs 6.0% renewables. The risk premium is a property of the *technology*, not the scenario.

> **Why higher returns mean a higher discount rate.** A higher expected return is compensation, and compensation is a discount rate. If investors will only hold a coal utility when it pays them 2% per year more than a comparable renewables firm, then by definition they are discounting the coal firm's future cash flows at a higher rate. Expected return and cost of equity are the same number seen from two sides — the investor's reward is the company's cost of capital.

Differential impact (iso_d1 vs vanilla, carbontech expected-negative %):

PROVIDER	VANILLA	ISO_D1	Δ (PP)
WITCH 5.0	75.3*	84.1	+8.8
MESSAGEix-GLOBIOM 1.1	85.3	87.4	+2.1
IMAGE 3.0	93.6	94.0	+0.4
COFFEE 1.1	52.8	53.0	+0.2
GCAM 5.2	52.2	52.0	−0.2

Greentech also improves (e.g. COFFEE 86.2→91.3% positive). Modest alone — but the isolation study identifies D1 as **the prime candidate behind the +19.5pp average interaction term** (§9): it is the component that converts MCPR's revenue lift into a net brown penalty.

Extended impact — pooled across the 5 providers (vanilla → iso_d1)

CLASS	BASELINE NPV>0	SHOCK NPV>0	EXPECTED DIRECTION	MEDIAN ΔNPV
High-carbon	57.5 → 58.2%	37.5 → 36.2%	70.5 → 72.8%	−55.8 → −60.4%
Low-carbon	97.9 → 97.9%	98.7 → 98.7%	89.3 → 90.4%	+90.4 → +97.3%

How to read these (same format in every box): Baseline/Shock NPV>0 = share of cells with a viable (positive-NPV) business in each scenario; Median ΔNPV = the typical shock size ((shock−baseline)/|baseline|) — for high-carbon, more negative = harsher shock, and the change vs vanilla is the update's effect on shock severity (the "difference of the difference"). Regional expected-direction, carbon: Advanced 66.6→70.0 · China 77.8→78.3 · Middle East 48.2→50.2. Green Middle East improves 93.4→97.8.

U2 Terminal-value package: stranding-aware TV, 3-yr normalization, 0%/2% growth split

Vintage: the impact figures in this box are from the April-2026 isolation study, on pre-rebuild data. They have not been re-measured against the 2026-09-01 inputs — that needs an `iso_*` sweep (3 configs × 25 providers, ~4.5h). Direction of effect is expected to hold; magnitudes will shift, since the rebuild lifted the vanilla baseline from 71.9% to 81.5%.

`commit 056d1f6` `config iso_stranding_tv (partial)`

The issue. Terminal value is ~15–20× final-year cash flow, so it dominates every NPV — and the December model gave every asset, declining coal included, a perpetuity growing at 2% off the raw final modeled year. Any distortion in that single year (a CapEx spike, one bad year) was capitalized forever, and fossil assets were implicitly assumed to operate and grow without end in a below-2°C world. Note the asymmetry: a bad year in the *middle* of the horizon costs only its own discounted cash flow, and an asset retiring mid-horizon correctly loses its terminal value (its final-year cash flow is ~0). The multiplier bug bites only when a one-off — a retirement wave, a decom charge, a CapEx spike — lands in the *anchor* year, where it gets turned into a phantom perpetual cost; transition scenarios concentrate exactly such events in the 2040s.

What changed. Three-tier terminal value: 3 consecutive loss-making final years → TV=0 (stranded); profitable-but-declining fossil → 10-year finite annuity; green → Gordon perpetuity. Terminal cash flow averaged over the last 3 years instead of final-year-only. Terminal growth split: fossil 0%, green 2%.

Source & intuition. Gourdel (2024) real-options abandonment logic; Damodaran/McKinsey normalization practice (one CapEx spike shouldn't erase a perpetuity); declining fossil assets have finite economic lives.

Differential impact (stranding TV isolated, carbontech Δpp vs vanilla — selected from the 25-provider study):

PROVIDER	Δ (PP)	PROVIDER	Δ (PP)
GEM-E3 V2021	+5.0	GCAM 5.2	−1.0
COFFEE 1.1	+1.0	IMACLIM 1.1	−1.4
AIM/CGE 2.2	−0.5	GCAM-PR 5.3	−3.4
25-provider average			**+0.6** (range −3.4 to +5.0)

Near-zero alone — most fossil cells aren't loss-making for 3 consecutive terminal years until the pricing and discount changes push them there. Normalization and the growth split were **not isolated**; their effect sits inside the interaction term. Honest gap to state.

Extended impact — pooled across the 5 providers (vanilla → iso_stranding_tv)

CLASS	BASELINE NPV>0	SHOCK NPV>0	EXPECTED DIRECTION	MEDIAN ΔNPV
High-carbon	57.5 → 58.5%	37.5 → 37.9%	70.5 → 71.1%	−55.8 → −54.0%
Low-carbon	97.9 → 97.9%	98.7 → 98.8%	89.3 → 90.0%	+90.4 → +88.5%

Regional expected-direction, carbon: Advanced 66.6→68.9 · China 77.8→74.3 (a regression — stranding bites Chinese coal baselines too) · Middle East 48.2→46.5.

U3 Retirement economics: decommissioning sign + no double-charging

Vintage: pre-study analysis from an earlier cell universe; never re-isolated (the fix entered before the study's vanilla was frozen, so its effect is embedded in every other table). Not re-measured on the 2026-09-01 inputs.

`commits 056d1f6, 31596a0` `not in the 125-run matrix`

The issue. Retirement economics were backwards twice over: retiring an asset *paid* the company (decommissioning was booked as scrap income), while the same retiring asset was still being charged replacement CapEx in its final year. Early retirement in a shock scenario — the central mechanism of transition risk — was being rewarded on one line and double-charged on another.

What changed. Decommissioning flipped from positive scrap income to a real cash *cost* (50% of CapEx/MW — demolition, remediation); retiring assets no longer also pay replacement CapEx in the same year (the 2% annual rollover applies to operating assets only).

Source & intuition. A stranded plant has no salvage market — it has ARO-style liabilities. EPRI/Lazard benchmark for the 2% maintenance rollover. Code review caught the double-charge.

Differential impact (dedicated pre-study analysis, earlier cell universe):

METRIC	BEFORE FIX	AFTER FIX	Δ (PP)
Carbontech expected-negative %	52	67	+15

The single largest lever found in that phase. It entered the code before the study's `vanilla` was frozen, so it is embedded in *all* five study configs and was never re-isolated. Flag: the 52→67 figure and the 68.5→78.5 headline use different cell universes — do not chain them.

Extended metrics not computable for this update: it entered before the study's `vanilla` was frozen, so no stored config isolates it — its effect is embedded in every column of every other table.

U4 Full-EF carbon costing mode

Vintage: the impact figures in this box are from the April-2026 isolation study, on pre-rebuild data. They have not been re-measured against the 2026-09-01 inputs — that needs an `iso_*` sweep (3 configs × 25 providers, ~4.5h). Direction of effect is expected to hold; magnitudes will shift, since the rebuild lifted the vanilla baseline from 71.9% to 81.5%.

`config iso_full_ef`

The issue. The original carbon-cost formula charged only emissions *above* the marginal generator's — which means the marginal technology itself, gas, paid no carbon cost at all. That is defensible when carbon costs are already embedded in the scenario's electricity prices, but in scenarios where they are not, it lets the most exposed bridge fuel escape carbon costs entirely.

What changed. Two carbon-cost modes: `full_ef` (every tonne pays — current default) vs `differential_ef` (only emissions above the marginal generator pay; for scenarios whose prices already embed carbon).

Differential impact (iso_full_ef vs vanilla, carbontech expected-negative %):

PROVIDER	VANILLA	ISO_FULL_EF	Δ (PP)
WITCH 5.0	75.3*	83.4	+8.1*
COFFEE 1.1	52.8	52.8	0.0
GCAM 5.2	52.2	52.2	0.0
IMAGE 3.0	93.6	93.6	0.0
MESSAGEix-GLOBIOM 1.1	85.3	85.3	0.0

Its real role is correctness — preventing carbon double-counting — rather than direction improvement; the auto-detection guard (U6) decides when it applies. (*WITCH vintage caveat, §9.)

Extended impact — pooled across the 5 providers (vanilla → iso_full_ef)

CLASS	BASELINE NPV>0	SHOCK NPV>0	EXPECTED DIRECTION	MEDIAN ΔNPV
High-carbon	57.5 → 57.5%	37.5 → 35.4%	70.5 → 72.2%	−55.8 → −62.0%
Low-carbon	97.9 → 97.9%	98.7 → 98.7%	89.3 → 89.3%	+90.4 → +90.4%

Charging gas its full emission factor deepens the typical carbon shock (−55.8→−62.0%) without touching baselines or green assets. Regional carbon direction: Advanced 66.6→69.0 · China 77.8→78.1 · Middle East 48.2→48.7.

9 The combined picture

> **Re-measured 2026-09-01 on fully rebuilt inputs.** Every number in this section comes from a 25-provider `vanilla` vs `suite_no_mcpr` sweep on source data with no reconstructions in the chain: BigQuery marts of 2026-09-01, ownership resolved via `ownership_type=='direct'`, the ×100 capacity-allocation bug fixed, and carbon prices corrected at source. It supersedes the April 125-run study figures (`68.5% → 78.5%`) that earlier versions of this document quoted.

Carbontech expected-negative %, by provider (vanilla → suite_no_mcpr)

PROVIDER	VANILLA	SUITE_NO_MCPR	Δ (PP)
AIM CGE 2.2	93.4	98.5	+5.1
COFFEE 1.1	43.3	79.5	+36.2
GCAM 5.2	54.6	75.5	+20.9
GCAM 5.3	60.8	76.8	+16.0
GCAM-PR 5.3	41.5	56.9	+15.4
GEM-E3 V2021	74.9	91.6	+16.7
IMACLIM 1.1	97.1	99.3	+2.2
IMAGE 3.0	96.4	99.0	+2.6
IMAGE 3.2	92.2	97.8	+5.6
MESSAGEix-GLOBIOM 1.1	66.5	84.0	+17.5
POLES ENGAGE	97.1	99.6	+2.5
POLES GECO2019	97.3	99.4	+2.1
REMIND 2.1	88.7	95.2	+6.5
REMIND-Buildings 2.0	84.8	92.3	+7.5
REMIND-H13 2.1	86.4	95.1	+8.7
REMIND-MAgPIE 1.7-3.0	97.1	98.1	+1.0

REMIND-MAgPIE 2.0-4.1	85.9	94.2	+8.3
REMIND-MAgPIE 2.1-4.2	85.3	96.4	+11.1
REMIND-MAgPIE 2.1-4.3	79.7	90.4	+10.7
REMIND-Transport 2.1	83.2	92.3	+9.1
TIAM-ECN 1.1	85.8	92.2	+6.4
TIAM-Grantham 3.2	82.4	89.2	+6.8
TIAM-UCL 4.1.1	93.8	99.1	+5.3
WITCH 5.0	88.1	96.4	+8.3

24 of 24 providers improve; mean +9.7pp. PyPSA-Eur-Sec 0.0.2 is excluded — its scenario carries no fossil technologies (hydro, nuclear and wind only), so carbontech direction is undefined rather than failing.

- **Headline:** carbontech direction moves 81.5% → 91.2% (+9.7pp) across 25 providers. Greentech gives back −2.7pp (64.3 → 61.6) — the same accepted trade-off, slightly larger.
- **MCPR contributes nothing.** `adjusted` (suite + MCPR) scores 91.2% — identical to `suite_no_mcpr` to the decimal. Dropping MCPR costs zero direction accuracy, now confirmed on completely rebuilt data.
- **OilCap remains the star correction:** 35.3% → 80.6% negative (+45.3pp). Coal 90.9 → 98.1; gas 78.7 → 86.5 — gas is still the structural borderline, but far less so than before.
- **The updates substitute for a missing carbon-price signal.** This is the strongest finding in the rebuild. Lift is inversely related to whether a scenario publishes carbon prices: providers that publish none gain most — COFFEE +36.2, GCAM 5.2 +20.9, GCAM 5.3 +16.0, GCAM-PR +15.4 — while providers that do publish them already sit near 97% under vanilla and barely move (REMIND-MAgPIE 1.7-3.0 +1.0, POLES GECO2019 +2.1, IMACLIM +2.2). Eight of the 25 providers publish no carbon price at all, so this is the common case, not the edge case. The updates buy *robustness to scenario data gaps*.
- **The lift is on marginal cells, not material ones.** Of 98,916 matched carbontech cells, 10,347 flip to the expected negative and 1,044 flip the other way — a 10:1 ratio. But the flipped cells have a median |ΔNPV| of 13.0% under vanilla, against 70.2% across all carbontech cells, and the median carbontech ΔNPV barely moves (−64.1 → −65.4%). The updates resolve the *sign* on borderline assets; they do not deepen the shock on assets that were already clearly exposed. State this plainly.

Extended impact of the suite — pooled across all 25 providers (vanilla → suite_no_mcpr)

CLASS	BASELINE NPV>0	SHOCK NPV>0	EXPECTED DIRECTION	MEDIAN ΔNPV
High-carbon	66.0 → 58.5%	50.5 → 45.3%	81.5 → 91.2%	−64.1 → −65.4%
Low-carbon	79.1 → 69.0%	78.8 → 70.6%	64.3 → 61.6%	+18.8 → +15.5%

The suite is not uniformly better, and the deck should say so. It lifts carbontech direction by 9.7pp, but baseline viability falls for both classes (high-carbon 66.0 → 58.5%, low-carbon 79.1 → 69.0%) and greentech direction slips 64.3 → 61.6%. D1 spreads and the terminal package weaken baselines as well as shocks. The median carbontech shock deepens only slightly (−64.1 → −65.4%), consistent with the marginal-cell finding above.

Regional picture — carbontech expected-negative % (pooled, 25 providers)

REGION	N (CARBON)	VANILLA	SUITE_NO_MCPR	Δ (PP)
China	35,561	92.3	97.2	+4.9
Advanced (EU/US/OECD)	33,042	76.2	91.0	+14.8
Middle East	4,730	57.3	74.4	+17.1

Advanced = EU, USA, CAN, JPN, KOR, R10EUROPE, R10NORTH_AM, R10PAC_OECD; China = CHN, R10CHINA+; Middle East = R10MIDDLE_EAST. **The Middle East conclusion has reversed.** Earlier versions of this document called it "the unsolved region" — barely half-right in vanilla and essentially unimproved by the suite (48.2 → 49.2). On rebuilt data it is the *largest* regional gain (57.3 → 74.4, +17.1pp). The earlier result was an artefact of the two data defects since fixed: ownership attributed to the wrong tier, and carbon prices frozen at their first year. It remains the weakest region in absolute terms, so an ME-specific price treatment is still worth discussing — but it is no longer evidence that the suite fails there.

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10 Caveats & open items for the discussion

- **Two vintages coexist in this document.** Section 9 is rebuilt (2026-09-01, all-source data, 25 providers). Sections U1–U4 are April-2026 isolation-study numbers. Do not chain a delta from one against a level from the other.
- **MCPR has been dropped** from the model and from this document. It contributed `+0.0`pp on rebuilt data, and its isolated behaviour was unstable across providers. The former U5–U7 boxes are removed; the empirical-MCPR question is a paper-track item, not a model dependency.

- **Never chain figures across vintages or cell universes.** The 52→67% decommissioning figure, the April study's 68.5→78.5% suite figure, and the rebuilt 81.5→91.2% in section 9 are three different measurements on three different cell universes. Quote section 9 for anything external.
- Terminal normalization, growth split, retirement double-charge fix, and the VRE-baseline bug fix have no isolated runs — their effects are embedded, stated qualitatively.
- **Ownership coverage.** 10.6% of asset-years sum to *under* 100% ownership (p01 = 25%), so a small slice of assets is only partly attributed. Worth raising upstream.
- **Not uniformly better.** The suite lifts carbontech direction but lowers baseline viability for both classes and greentech direction by 2.7pp. Present it as a sharpening of carbon discrimination, not a strict improvement.

A Appendix — evidence base & metric definition

Direction metric. For each company-technology cell: does the NPV change between baseline and shock have the economically expected sign? Carbontech (coal, gas, oil) should be *negative*; greentech (wind, solar, hydro) should be *positive*. "Carbon neg%" = share of carbontech cells with the expected negative sign.

Data sources, all already in the repository:

- **125-run isolation study** (2026-04-12): 25 IAM providers × 5 configs — `vanilla` (all adjustments off = December-2025 behavior on current plumbing), `adjusted` (all on), `iso_mcpr`, `iso_stranding_tv`, `iso_dynamic_ef`. 828,791 company-technology records. Source: `workspace/comparison_results/COMPARISON_REPORT.md`.
- **Rebuild sweep** (2026-09-01): 25 providers × `vanilla` / `suite_no_mcpr`, plus `adjusted` for the MCPR check — on all-source data. This is the basis for section 9.
- **Fresh per-config recomputation** (2026-07-22): direction rates recalculated directly from the stored `company_technology_npv.csv` files for the tables above.

Sources: `workspace/comparison_results/COMPARISON_REPORT.md`, `summary_table.csv`, `workspace/scenario_manifest.json`, `discover_scenario_pairs.py`, per-config `company_technology_npv.csv` recomputation (2026-07-22), commits `056d1f6` – `7b7546a`, plan of record `docs/superpowers/plans/2026-07-06-altr-reconciliation-and-effectiveness-plan.md`. Companion docs: `altr_updates_Brief_v1.html`, `altr_improvements_Deck_v2`.